MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 3

1. *Find the inverses of the following invertible matrices:*

(1) $\begin{bmatrix} 8 & 6 \\ 5 & 4 \end{bmatrix}$

(2) $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

2. *Is the sum of two invertible matrices always an invertible matrix? Justify your answer.*

3. *If a matrix A is invertible, is the matrix A^2 invertible? Justify your answer.*

4. *Let A be an invertible $n \times n$-matrix and $\mathbf{b}$ be a vector in $\mathbb{R}^n$. How many solution does $A\mathbf{x} = \mathbf{b}$ have? Explain why!*

Solutions will be posted at the end of Week 4.